(1) Shankar Rao page 97

Second order PDE with variable coefficients:

A PDE is of the form

$$F(xD_x, yD_y) = f(x,y)$$

Let $x = e^u$, $y = e^v$

$\therefore u = \ln x$ $\quad v = \ln y$

$\boxed{\frac{\partial u}{\partial x} = \frac{1}{x}}$

$$D_x z = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} = \frac{1}{x}\frac{\partial z}{\partial u} \quad \therefore \boxed{x D_x z = D_u z}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{1}{x}\frac{\partial z}{\partial u}\right) = \frac{\partial}{\partial u}\left(\frac{1}{x}\frac{\partial z}{\partial u}\right)\frac{\partial u}{\partial x}$$

$$= \frac{1}{x}\frac{\partial}{\partial u}\left(\frac{1}{x}\frac{\partial z}{\partial u}\right)$$

$$= \frac{1}{x}\left(\frac{1}{x}\frac{\partial^2 z}{\partial u^2} - \frac{\partial z}{\partial u}\cdot\frac{1}{x^2}\frac{\partial x}{\partial u}\right)$$

$$= \frac{1}{x}\left(\frac{1}{x}\frac{\partial^2 z}{\partial u^2} - \frac{\partial z}{\partial u}\cdot\frac{1}{x^2}\cdot x\right) = \frac{1}{x^2}\left(\frac{\partial^2 z}{\partial u^2} - \frac{\partial z}{\partial u}\right)$$

$$\therefore \boxed{x^2\frac{\partial^2 z}{\partial x^2} = D_u(D_u - 1)z}$$

Similarly, $x^3\frac{\partial^3 z}{\partial x^3} = D_u(D_u-1)(D_u-2)z$

Proceeding in a similar way, we get

$$\boxed{yD_y z = D_v z} \quad \boxed{y^2 D_y^2 z = D_v(D_v-1)z}$$

$$\boxed{y^3 D_y^3 z = D_v(D_v-1)(D_v-2)z}$$

Ex. 1;

$$x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = x^2 y \quad (1)$$

Let

$$x = e^u, \quad y = e^v$$

$$\Rightarrow u = \ln x \quad | \quad v = \ln y$$

$$(1) \Rightarrow \left\{ D_u(D_u - 1) - D_v(D_v - 1) \right\} z = e^{2u} \cdot e^{v} = e^{2u+v}$$

$$\Rightarrow \left(D_u^2 - D_u + D_v - D_v^2 \right) z = e^{2u+v}$$

A.E, $m^2 - m + \cancel{1} - \cancel{1} = 0$

$$m(m-1) = 0 \therefore m = 0, \; m = +1$$

$$z_c = \phi_1(v) + \phi_2(v+u) = \phi_1(\ln y) + \phi_2(\ln y + \ln x)$$

$$z_p = \frac{1}{D_u^2 - D_u + D_v - D_v^2} e^{2u+v}$$

$$= \frac{1}{4 - 2 + \cancel{1} - \cancel{1}} e^{2u+v}$$

$$= \frac{1}{2} e^{2u+v} = \frac{1}{2} e^{2u} \cdot e^{v}$$

$$= \frac{1}{2} x^2 \cdot y$$

$$\therefore \boxed{z = z_c + z_p}$$

3 $\left(x^2\frac{\partial^2 z}{\partial x^2} - 4y^2\frac{\partial^2 z}{\partial y^2} - 4y\frac{\partial z}{\partial y} - z\right) = x^2y^3\ln y$ —— (1)

Let $x = e^u, \quad y = e^v$

$u = \ln x, \quad v = \ln y$

$\therefore \left[D_u(D_u-1) - 4D_v(D_v-1) - 4D_v - 1\right]z = e^{2u}\cdot e^{3v}\cdot v$

$\Rightarrow \left(D_u^2 - 4D_v^2 - D_u - 1\right)z = v\,e^{2u+3v}$ —— (2)

A.E, $m^2 - 4\cdot 1 - m - 1 = 0$

$\Rightarrow m^2 - m - 5 = 0$

$m = \frac{1 \pm \sqrt{1-4\times 5}}{2\cdot 1}$

$= \frac{1 \pm \sqrt{19}\,i}{2} = \frac{1}{2} \pm i\frac{\sqrt{19}}{2}$

$z_c = \phi_1\left(v + \left(\frac{1}{2} + \frac{\sqrt{19}}{2}i\right)u\right) + \phi_2\left(v + \left(\frac{1}{2} - \frac{i\sqrt{19}}{2}\right)u\right)$

$= \phi_1\left\{\ln y + \left(\frac{1}{2} + \frac{\sqrt{19}}{2}i\right)\ln x\right\} + \phi_2\left(\ln y + \left(\frac{1}{2} - \frac{i\sqrt{19}}{2}\right)\ln x\right)$

(2) $\Rightarrow z_p = \frac{1}{D_u^2 - 4D_v^2 - D_u - 1}\, v\, e^{2u+3v}$.

$= e^{2u+3v}\,\frac{1}{(D_u+2)^2 - 4(D_v+3)^2 - (D_u+2) - 1}\, v$

$$z_p = e^{2u+3v} \frac{1}{D_u^2 - 4D_v^2 + 3D_u - 24D_u - 35} v$$

$$= e^{2u+3v} \frac{1}{-35} \left(1 + \frac{24 D_v}{35}\right)^{-1} v$$

$$= e^{2u+3v} \left(-\frac{1}{35}\right) \left(1 - \frac{24}{35} D_v + \cdots\right) v$$

$$= -\frac{1}{35} e^{2u+3v} \left(v - \frac{24}{35}\right)$$

$$= -\frac{1}{35} e^{2u} \cdot e^{3v} \left(\ln y - \frac{24}{35}\right)$$

$$= -\frac{1}{35} x^2 y^3 \left(\ln y - \frac{24}{35}\right)$$

$$\therefore \boxed{z = z_c + z_p}$$

$$\boxed{\frac{\partial^2 z}{\partial x \partial y} = D_x D_y}$$

Ex: Solve

$$\left(x^2 \frac{\partial^2 z}{\partial x^2} - 2xy \frac{\partial^2 z}{\partial x \partial y} - 3y^2 \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial x} - 3\frac{\partial z}{\partial y}\right) = x^2 y \sin(\ln x^2) \quad \text{--- (1)}$$

Let $x = e^u$, $y = e^v$

$u = \ln x$ $\quad v = \ln y$

$$\Rightarrow \left[D_u(D_u - 1) - 2 D_u D_v - 3 D_v(D_v - 1) + D_u - 3D_v\right] z = e^u \cdot e^v \sin(2u)$$

$$= e^{u+v} \sin(2u)$$

$$\Rightarrow \left(D_u^2 - 2 D_u D_v - 3 D_v^2\right) z = e^{2u+v} \sin(2u) \quad \text{--- (2)}$$

A.E, $m^2 - 2m - 3 = 0$

$\therefore m = -1, 3$

$$z_c = \phi_1(v - u) + \phi_2(v + 3u) = \phi_1(\ln y - \ln x) + \phi_2(\ln y + 3\ln x).$$

$$z_p = \frac{1}{D_u^2 - 2D_u D_v - 3D_v^2} e^{2u+v} \sin 2u$$

$$= -\frac{x^2 y}{65}\left[4\cos 2u + 7\sin 2u\right]$$

$$= \frac{-x^2 y}{65} 4\left[\cos(2\ln x) + 7\sin(2\ln x)\right]$$

MKT

Ex: 11.25

* $(x^2 D_x^2 + 2xy\, D_x D_y + y^2 D_y^2) z = x^2 y^2$

$\cancel{z_c = \phi_1(\ln x - \ln y) + x\phi_2 \ln x}$

* $(D_x^2 + D_y^2) z = -x(x^2 + y^2)$